Ng Au Hern Wesley

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Education

Singapore University of Technology and Design — Singapore
Bachelor of Engineering, Computer Engineering — May 2026
Global Merit Scholarship
Engineering Product and Development

University of California, Berkeley — California, USA
Berkeley Summer Sessions — Jun 2023 – Aug 2023
Completed coursework in Game Theory and Media Studies

Hwa Chong Institution — Singapore
Integrated Programme — Nov 2019
Hwa Chong Diploma with Merit
Rank Points of 86.25 (Computing, Physics, Math and Economics)

Technical Skills & Interests

- **Systems & Environments**: Linux (Ubuntu), SSH, Git, ROS 2
- **Programming & Scripting**: Python, C++, Scala
- **Hardware & Design**: Hardware telemetry, MCU development, Fusion 360, 3D Printing
- **Language**: Fluent in English and Mandarin (spoken and written)
- **Interests**: Sports, outdoor activities, video essays, gaming

Research & Projects

Capstone Project, SUTD - SITA — Sept 2025 – May 2026
Autonomous Luggage Concierge

- Developed B.HIVE, an autonomous Baggage Handling Intelligent Vehicle Ecosystem for SITA, designed to assist travelers with mobility challenges by providing a hands-free baggage transport solution
- Engineered the robot's autonomous navigation and dynamic obstacle avoidance capabilities by utilising a Jetson Orin Nano and performing sensor fusion with LiDAR and depth cameras
- Integrated a Bluetooth-based smart following algorithm paired with a custom Progressive Web App (PWA), allowing the robot to actively track the user and dynamically adjust to their walking pace
- Constructed a scaled functional prototype featuring a custom chassis, integrated motor controllers, and a baggage weighing mechanism, validating SLAM mapping and navigation through RViz simulations and physical testing

Digital Systems Lab, SUTD — Jan 2025 – May 2025
Cryptographically Secure Random Number Generator (CSRNG) utilizing Environmental Radiation implemented using FPGA Platform

- Designed a CSPRNG using FPGA hardware and Geiger counter inputs to establish robust hardware-level security and entropy extraction for cryptographic key generation
- Evaluated generated cryptographic keys against rigorous standard statistical test suites to validate data integrity and true randomness
- Co-authored and published technical findings on IEEE, detailing the hardware architecture and statistical validation of the generated cryptographic keys

Dyson-SUTD Undergraduate Research Opportunities Programme April 2024 – Dec 2024

Electrostatic Discharge (ESD) Characterisation and Analysis for Human-Device Interaction

- Investigated Electrostatic Discharge (ESD) phenomena in cordless vacuum cleaners to analyze human-device interaction and understand how static charges build up on non-conductive surfaces
- Designed and executed physical testing methods, utilizing polystyrene bead dispersion to successfully model and visualize triboelectric charge distribution across the device
- Utilized a Van de Graaff generator to artificially charge the device and a coulombmeter to quantify charge propagation

Experience

Defence Science and Technology Agency Aug 2024 – Dec 2024

Land Systems, Safe Deployment of Robots in Mixed Environments (Smart Workshop)

- Internship under DSTA Land Systems Programme Center
- Developed a real-time battery temperature telemetry sensor to detect thermal runaway anomalies, ensuring critical hardware safety and operational continuity.
- Researched how safety can be improved during deployment of Autonomous Mobile Robots (AMRs) within Mixed Environments (Spaces with active moving personnel and vehicles) in Army camps
- Published a literary review, putting focus on perceived safety alongside physical safety, while analysing relevant safety studies

Republic of Singapore Navy Mar 2020 – Jan 2022

Naval Warfare Systems Operator

- Executed complex ammunition logistics for the RSN, obtaining both military and commercial forklift licenses to ensure safe and efficient operations
- Selected to represent Naval Logistics as a Colours Party Orderly during the National Day Parade

Leadership & Activities

DSTA Technology Undergraduate Programme Jan 2024 – May 2026

Member

- Offered internship and courses conducted by DSTA, in relation to the defence sector of Singapore
- Obtained certification for the following courses: Cyber Fundamentals, MINDEFSAF on Commercial Cloud, Wireless Communications

Singapore University of Technology and Design Nov 2023 – Feb 2024

Logistics Director, Open House Committee

- Coordinated logistics for over 80 Fifth Rows (Co-curricular Activities)
- Designed floor plan allocation to ensure smooth visitor journey

Logistics Executive, Freshmen Orientation Committee Mar 2023 – Sept 2023

- Awarded Student Service Award at Student Achievement Awards 2024
- Prepared and acquired logistics to ensure smooth activity planning by other departments